S06. Inverse pair-potential design over an explicit competitor class

Pure Thought research portfolio

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Source: synthesis challenge 20; original synthesis. This topic is distinct from, or adds a distinct objective to, the canonical PRD catalog. Crosswalk

Plain-language guide

The problem in everyday terms

Instead of asking which arrangement particles prefer under known forces, turn the question around: which interaction would favor an arrangement we want? Here, candidate pair interactions are tuned so that a target repeating structure has lower energy than an explicitly listed set of alternatives across a range of densities.

Key terms

- **Pair potential** — A function giving the interaction energy of two particles as a function of their separation.
- **Inverse design** — Choosing model parameters to produce a desired outcome.
- **Target lattice** — The repeating particle arrangement the design aims to favor.
- **Competitor class** — The precise collection of alternative arrangements against which the target is compared.
- **Number density** — The number of particles per unit volume or area.
- **Energy per particle** — Total interaction energy normalized by the number of particles in the periodic limit.
- **Lattice sum** — A sum of interactions over particles in a repeating structure.
- **Tail bound** — A guaranteed bound on contributions omitted when truncating an infinite sum.
- **Ground state** — A lowest-energy configuration within a stated admissible class.

Why this matters

Interaction design is a way to explore self-assembly: simple local rules may favor organized structures. Reliable comparisons expose whether apparent success comes from a useful mechanism or from overlooking a competing arrangement or a long-range contribution.

What progress would mean

Progress would prove the target's energy advantage, or identify a defeating competitor, within the declared family and density interval. Beating that family alone does not establish the ground state among every possible configuration or show that particles will assemble into it at finite temperature.

Revised core question

Can a parameterized pair potential make a target periodic structure an energy minimizer among a precisely specified set of competitors over a density interval?

Precise scope, target and outputs

Fix dimension, density interval, and v_θ(r)=Σ_jθ_jϕ_j(r) with bounded coefficients and convergent lattice sums. Define E_θ(Λ)=1/2 Σ_{x$\in\Lambda\setminus\{0\}$}v_θ(|x|). Begin with a finite list of normalized lattice competitors and use interval sums plus tail bounds to certify E_target$\leq$E_competitor$-\delta$ uniformly on a density interval. Continuous competitors require a compact parameterization and global coverage. State whether particle displacements or variable cell size are allowed. General configuration ground states require additional universal lower bounds beyond lattice comparisons.

Detailed critique of the synthesis framing

The original jumps from optimizing lattice energies to provable crystals over both densities and temperatures. Beating selected lattices does not exclude aperiodic competitors or infinitesimal deformations, and a zero-temperature energy gap is not a finite-temperature free-energy theorem. A strictly uniform energy gap against all arbitrarily close deformations may be impossible under continuity. The rewrite gives a restricted inverse-design result whose proof scope is honest; unrestricted crystallization and thermal stability remain separate research extensions.

Validation and milestones

Baseline: exact or interval-verified ordering for a finite competitor list. Strong: uniform density/parameter-region certificate. Validate a direct long-cutoff sum against the certified tail bound; label the result "best among these competitors" unless a universal proof is supplied.

Use the shared evidence standard. Unsuccessful search is unresolved unless a complete exclusion proof is supplied. These are proposed baseline specifications, not completed scientific results.

Displayed formulation and claim boundary

Supplemental specification.

$$v_\theta(r) = \sum_j \theta_j \phi_j(r), \qquad E_\theta(\Lambda) = \frac{1}{2} \sum_{x \in \Lambda \setminus \{0\}} v_\theta(|x|),$$

$$E_\theta(\Lambda_{\text{target}}) \leq E_\theta(\Lambda_{\text{competitor}}) - \delta.$$

Require convergent sums and certified tails, uniformly on the declared density interval and competitor set. The conclusion is best among those competitors. Unrestricted configurations, infinitesimal deformations and finite-temperature free energies require separate arguments.

Evidence and execution standard

This standard applies to the revised specifications. It is a proposed contract for future research runs; this editorial revision does not supply the scientific certificates themselves.

What "pure thought" means here

Use explicitly stated mathematical models and deductions without fitting conclusions to experimental outcomes. Synthetic examples, exact model inputs, analytic references, numerical search, and established software are allowed. Physical constants, electronic-structure approximations, constitutive laws, and empirical parameter sets must be identified when used. A model-derived result is conditional on that model.

The original ban on legacy software conflicts with its proposed reliance on NumPy, SciPy, CVXPY, nauty, and other libraries. Reimplementing every optimizer and algebra system is not necessary for scientific independence. Prefer a small independently checkable certificate and a documented trusted base. A fresh implementation can be a project objective, but must not be confused with a mathematical proof or novelty claim.

Four evidence levels

Label	What it establishes	What it does not establish
Numerical evidence	A reproducible computation at stated precision, sample size, and discretization	Exact truth, continuum coverage, or a universal theorem
Finite-model certificate	An exact/validated statement about the specified finite model or finite relaxation	An omitted tail, thermodynamic limit, or physical realization
Conditional mathematical theorem	A proof over the stated domain with every hypothesis and error controlled	Truth outside those assumptions or a demonstrated physical system
Formal proof	A checked theorem in a named proof assistant, with imported axioms/dependencies listed	Correct physical modeling merely by virtue of formalization

Formalization is an optional extra deliverable unless the specific project makes it mandatory. High precision, JSON output, solver status and small residuals do not by themselves confer any of these proof levels.

Claim directions

For minimization over a larger feasible set, the optimum is a lower bound on the original minimum. A physical feasible point supplies an upper bound. For maximization the directions reverse. Track feasibility of the primal/dual witnesses separately from their objective values.

Infeasibility of an inner ansatz excludes only that ansatz. Infeasibility of an outer relaxation can exclude the original problem if the map into the relaxation is proved and all approximations preserve necessity. Passing necessary tests means "not excluded"; existence needs a constructive sufficient argument.

A smallest observed example is not a minimality theorem. A complete catalog must define the universe, equivalence relation, boundaries, and proof that pruning misses nothing. A finite-size numerical trend does not prove an asymptotic limit.

Minimum run manifest

Before calculation, record the problem ID and revision hash, exact inputs and conventions, quantified domain, equivalence relation, target observable, accepted evidence level, tolerances, and resource limits. Record dependencies and precision/rounding mode. Where randomization is used, preserve seeds, trial counts and raw sufficient statistics. Resource limits are planning decisions, not evidence that a mathematical solution cannot exist.

An output bundle should include:

- The statement being claimed, including all assumptions and quantifiers.
- Construction or proof object, plus an independently runnable verifier and its trust assumptions.
- Numerical/error ledger separating roundoff, discretization, truncation, optimization, sampling and modeling error where relevant.
- Positive reference cases and negative controls that the verifier must reject.
- Coverage: included domain, omitted domain, unresolved regions and termination reason.

These are requirements for future implementations, not a promise of polynomial-size certificates for every problem.

Checking specific certificate types

LP/SDP/SOS: verify equality constraints, objective direction, and cone feasibility using exact arithmetic or outward-rounded intervals. Rationalizing a numerical solution can break PSD constraints. Correct residuals with justified norm bounds; do not silently project and reuse the old objective. Distinguish a proof of a polynomial approximation from a proof for the original function.

Interval numerics: state function enclosures, domains and outward rounding. A small root box needs existence and, if claimed, uniqueness. Cover the complement for a global enumeration. Include singularities and tails rather than discarding them numerically.

SAT/SMT: a solver's UNSAT label is not a certificate. Preserve a proof supported by a checker and the proof that the encoding, symmetry reduction and blocked solutions match the mathematical universe. DRAT applies to an appropriate Boolean CNF proof, not automatically to every real-arithmetic or SDP claim.

Statistics: specify the estimand, dependence assumptions, confidence procedure and bias. Zero observed failures does not prove zero failure probability. MCMC error bars based on estimated autocorrelation are not generally rigorous mixing guarantees.

Milestones and novelty

Baseline means a working, independently validated result on a concrete reference problem. Strong means a broader or sharper result with its claimed evidence level. Research extension means a possible novel result whose feasibility and significance need review. A negative theorem can be valuable; an unsuccessful search can also be useful if its scope is documented, but the two are not interchangeable.

Do not promise "publication-quality" in a fixed number of months. Before claiming novelty, conduct a focused current literature comparison against the exact theorem/model/metric. The sources in

this revision check key definitions and limitations; they are not an exhaustive novelty audit across all 30 fields.